

Guidelines for Reporting About Network Data (GRAND)

These guidelines for reporting about network data describe the minimum information that should always be reported about a network dataset in a manuscript. They should be used to describe the final network data that are analyzed, following any transformations.

- Underlined terms are defined in the glossary. The glossary is provided to ensure clarity in the guidelines, however GRAND remains agnostic about terminology.
- **Bold** terms note guidelines for specific types of networks. Further adaptation may be necessary for special cases, or for types of networks not yet covered by GRAND.
- *Italicized* guidelines are optional, but describe additional information that may often be useful to report, particularly if the network is not described in more detail elsewhere.

1.1 Description: Network

Item	What to report
Type	Report the type of network: <u>Whole</u> , <u>Two-mode</u> , <u>Hypergraph</u> , <u>Multiplex</u> , <u>Multiwave</u> , <u>Multinet</u> , <u>Ego</u>
Source	Report the source of the network data: <u>Empirical</u> , <u>Generated</u>
Count	Multiplex: Report the number of relationships. Multiwave: Report the number of waves. Multinet: Report the number of networks. Ego: Report the number of ego networks. If count < 5, report the exact values of items below for each network, relationship, or wave. If count ≥ 5, report the range.
System	Report the system, location, and date that the network represents, at an ethically appropriate level of detail (e.g., an American high school in 2005). Ego: Report the population of egos, location, and dates that the ego networks represent, at a ethically appropriate level of detail (e.g., Mexican immigrants in New York in 2005).
Data	Report how to obtain the data, or why the data are not available.
Citation	<i>Provide a citation to a source where the data is more fully described.</i>

1.2 Description: Nodes

Item	What to report
Meaning	Report what the <u>nodes</u> represent using a word or phrase (e.g., students). Two-mode: Report what each type of node represents (e.g., researchers and articles).
Count	Report the number of nodes. Two-mode: Report the number of each type of node. Multiwave: Report (the range of) the number of nodes present in each wave. Multinet: Report (the range of) the number of nodes in each network. Ego: Report (the range of) the number of alters in each ego network.
Attribute	If the nodes have non-structural attributes (e.g., demographic characteristics), report the node attributes. <i>Describe how the attributes were measured, and provide sample descriptive statistics for each.</i>

1.3 Description: Edges

Item	What to report
Direction	Report whether the <u>edges</u> are undirected or directed.
Weight	Report whether the edges are unweighted, weighted, or signed.
Structural	If the network contains <u>structural edges</u> , report their presence and meaning.
Meaning	Report what the edges represent using a word or phrase (e.g., friendship). Multiplex: Report what each type of edge represents. Directed: Report the meaning of the edge direction (e.g., from origin to destination). Weighted: Report the meaning (frequency, similarity, intensity, count, or valence), level of measurement (ordinal or continuous), and range of weights.
Count	Report the number of edges (ignore weights, count directed edges separately). Multiplex: Report (the range of) the number of edges of each type. Signed: Report the number of positive edges and the number of negative edges. Multiwave, Multinet, Ego: Not applicable.

2.1 Data: Procedure

Item	What to report
Method	Empirical: Report the method used to collect the data using a word or phrase (e.g., survey, interview, observation, archival). <i>Provide a detailed description of the data collection methods, including (as applicable): how the network was bounded, how participants were recruited, the modality and procedures of data collection, and exact wording of question used to collect data.</i> Generated: Report the name and citation of the data generating model (e.g., <code>sample_smallworld()</code> in <code>igraph v2.1.4</code>) and the values of the function parameters. <i>Provide a detailed description of the data generating model, including its theoretical foundation and the role of each parameter.</i>
Ethics	Empirical: Report whether the data collection received institutional approval, and if so, the institution and approval date. <i>Provide a detailed description of any institutional approvals, procedures for obtaining informed consent, potential risks, and data security practices such as deidentification.</i>

2.2 Data: Error

Item	What to report
Degree	If a node's minimum or maximum degree is constrained by the data collection or generation design, report the restriction on degree (e.g., respondents could name up to 5 contacts).
Nodes	If any nodes within the system boundary are missing from the network, report the percent or number of missing nodes. Two-mode: ...report the percent or number of each type of missing node. Multinet: ...report (the range of) the percent or number of missing nodes in each network. Ego: Not applicable.

2.3 Data: Transformation

Item	What to report
Symmetrizing	If directed edges were transformed into undirected edges, report the transformation method (e.g., confirmed edges, unconfirmed edges). <i>Provide a detailed description of the transformation method.</i>
Binarizing	If weighted edges were transformed into unweighted edges, report the transformation method (e.g., threshold, backbone model). <i>Provide a detailed description of the transformation method.</i>
Projecting	If the network was generated from two-mode or non-network data such that edges represent similarities or co-occurrences, report the transformation method (e.g., projection, gaussian graphical model). <i>Provide a detailed description of the transformation method.</i>
Excluding	If any nodes have been excluded from the network (e.g., <u>isolates</u> , smaller components), report the exclusion criteria and percent of all nodes that were excluded. <i>Provide a detailed rationale for the exclusion of these nodes, and the analytic implications of their exclusion.</i>
Imputing	If any missing nodes or edges have been imputed, report the imputation method. <i>Provide a detailed description of the imputation method.</i>

3 Structure

Item	What to report
Components	Report the number of strongly-connected components. Multiplex: Report (the range of) the number of strongly-connected components for each type of edge. Multiwave: Report (the range of) the number of strongly-connected components in each wave. Multinet: Report (the range of) the number of strongly-connected components for each network. Ego: Not applicable.
Connectivity	Report either the <u>edge density</u> or mean degree, ignoring edge weights if present. Two-mode: Report the mean degree for each type of node. Hypergraph: Report the mean degree for nodes and the mean size for edges. Multiplex: Report (the range of) the edge density or mean degree for each type of edge. Multiwave: Report (the range of) the edge density or mean degree in each wave. Multinet: Report (the range of) the edge density or mean degree for each network. Ego: Report (the range of) the edge density for each ego network, and whether these values were computed with or without ego. Signed: Report edge density or mean degree separately for positive and negative edges.
Reciprocity	Directed: Report the percent of connected dyads that have a reciprocal edge, ignoring edge weights if present. Multiplex: Report (the range of) the percent of connected dyads that have a reciprocal edge for each type of edge. Multiwave: Report (the range of) the percent of connected dyads that have a reciprocal edge in each wave. Multinet: Report (the range of) the percent of connected dyads that have a reciprocal edge in each network. Ego: Report (the range of) the percent of connected dyads that have a reciprocal edge in each ego network.
Balance	Signed: Report the percent of connected triads that are weakly balanced (i.e., the triangle index). Multiplex: Report (the range of) the percent of connected triads that are weakly balanced (i.e., the triangle index) for each type of edge. Multiwave: Report (the range of) the percent of connected triads that are weakly balanced (i.e., the triangle index) in each wave. Multinet: Report (the range of) the percent of connected triads that are weakly balanced (i.e., the triangle index) in each network. Ego: Report (the range of) the percent of connected triads that are weakly balanced (i.e., the triangle index) in each ego network.
Similarity	Multiplex: Report (the range of) the jaccard similarity coefficient for each pair of edge types. Multiwave: Report (the range of) the jaccard similarity coefficient for each pair of consecutive waves.

Glossary: Network Types

Multiple types of networks exist, and in some cases require unique reporting guidelines. GRAND distinguishes different types of networks in terms of (1) the number of different entities that nodes represent, (2) the number of different relationships that edges represent, (3) what an edge can connect, and (4) the count of networks when a single dataset being described by GRAND is composed of multiple distinct networks.

Type	Entities	Relationships	Connect	Count	Synonyms	Example: A network describing...
Whole	one	one	dyads	one	complete one-mode sociocentric unipartite	Students connected by friendship
Two-mode	two	one	dyads of different entities	one	bipartite	Researchers connected to articles by authorship
Hypergraph	one	one	multiple nodes	one	higher order	Two or more researchers connected by authorship of the same an article
Multiplex	one	many	dyads	each relationship	–	Students connected by friendship; Students connected by bullying
Multiwave	one	one	dyads	each wave	temporal dynamic	Students connected by friendship in January; Students connected by friendship in February
Multinet	one	one	dyads	each network	–	Students in classroom A connected by friendship; Students in a classroom B connected by friendship
Ego	one	one	dyads	each ego	personal egocentric	A focal person and their sources of social support connected by support provision
<u>For inclusion in future versions of GRAND</u>						
Multimode	many	many	dyads of different entities	one	–	Researchers connected to articles by authorship, articles connected to journals by publication, and researchers connected to journals by editorship
Multilevel	many	many	dyads	one	–	Students connected by friendship, teachers connected by advice seeking, and students connected to teachers by classroom membership
Continuous	one	one	dyads	???	temporal dynamic	Students connected by speaking interactions that have specific start and stop times

Glossary: Terminology

Multiple terms often exist for a single network concept. To ensure clarity, this glossary defines the terms used in the reporting guidelines. However, GRAND remains agnostic about which term to use when synonyms exist.

Term (<i>possible synonyms</i>)	Definition
Edge (<i>link, tie, arc, connection, relationship</i>)	The relationships that connect nodes in a network, and whose arrangement gives a network its structure.
Edge density (<i>Density</i>)	Number of present edges divided by the number of possible edges, counting each present/possible directed edge separately.
Empirical	Data that are derived from the real world, directly or indirectly.
Generated	Data that are produced by a model or simulation.
Mean degree	Average degree over all nodes in a network, including any isolates.
Network	A graph where the nodes and edges are intended to represent something, and are not solely an abstract mathematical object.
Node (<i>vertex</i>)	The entities whose relationships are captured by a network.
Isolate	A node with degree of zero
Strongly-connected component	A set of nodes for which a path exists from every node to every other node. An isolate is defined as a component.
Structural Edges (<i>prohibited edges, required edges, structural zeros</i>)	A structural edge is an edge that either cannot exist, or must exist, <i>in principle</i> .